

US oceanographer Henry Stommel combined his sharp intellect with a passion for the sea to transform his discipline into a major scientific field. He helped to explain the powerful forces that drive ocean currents, including the Gulf Stream.

MILESTONES

FLUID DYNAMICS

Produces key paper, "The westward intensification of wind-driven ocean currents," in 1948.

SEAWATER SAMPLING

Sets up the PANULIRUS research station in 1954 in Bermuda, which gathers valuable and extensive data.

BEST-KNOWN TEXT

Publishes his book *The Gulf Stream* in 1958, having studied the current's global context for a decade.

ACADEMIC INTERLUDE

In 1959, becomes professor of oceanography at Harvard and publishes early theories on thermohaline effects.

NATIONAL MEDAL

Awarded the United States National Medal of Science in 1989 for his work on the physics of ocean currents.

Henry Stommel studied astronomy at Yale University, and taught there for 2 more years before joining the Woods Hole Oceanographic Institution (WHOI) in 1944. He later took up oceanography posts at Harvard and at the Massachusetts Institute of Technology (MIT), but returned to WHOI before the end of his career.

In 1948, Stommel published a groundbreaking paper showing how a combination of fluid dynamics, the Earth's curvature, and the Coriolis effect (in which moving objects are deflected to one side by the rotation of the Earth) causes stronger currents to be found on the western boundaries of ocean basins. His elegant solution explained the position and intensity of the Gulf Stream—a powerful, warm ocean current that runs up the western side of the Atlantic. The paper was characteristic of Stommel's intuitive grasp of complex phenomena, and it opened up a whole new area of study: the dynamics of large-scale ocean circulation.

Stommel and his colleagues developed his ideas into a wider theory of global ocean circulation with their work on thermohaline (temperature and salinity) effects. When sea ice forms at the poles, the seawater gets saltier and denser and sinks to the depths, pulling in warmer water behind it to create a conveyor belt of moving water looping through the world's oceans. Surface-level currents are predominantly wind-driven, but thermohaline forces drive deep-ocean circulation. Stommel later oversaw two of the most successful global studies of ocean processes.

The "Great Ocean Conveyor Belt," driven by variations in water temperature and salinity, is part of the global ocean circulation system envisioned by Stommel.

